

marketplace. Such arrangements are enabled through the station's unified docking system that will allow any commercial crew and cargo capability developed in the US, Europe and China.

It will be a hub for commercial activity, scientific research and development in low Earth orbit. Orbital Technologies has several customers already under contract

from different segments of industry and the scientific community, representing such areas as medical research and protein crystallization, materials processing, and the geographic imaging and remote sensing industry. Because of its design and its orbital inclination, the CSS will serve as an emergency refuge for the ISS crew, should it become necessary.

Finally, the CSS will be a true gateway to the rest of the solar system. A short stop-over at the station will be the perfect beginning of a manned circumlunar flight, says Orbital Technologies, adding that the deep space manned exploration missions planned in the next decade would be welcome to use the CSS as a waypoint and a supply station.

- **Bigelow Commercial Space Station** – While Orbital Technologies doesn't have any actual technology to show for their efforts, Bigelow Aerospace has actually achieved quite a bit, including launching a couple of satellites using their patented inflatable module.

Inflatable systems offer the potential for a greater amount of on-orbit volume while, relative to a traditional metallic structure, taking up a small amount

of rocket fairing space. Additionally, and perhaps most relevant to long-term orbital use, is the enhanced protection from radiation offered by inflatable habitats. Specifically, when exposed to cosmic rays or solar flares, traditional metallic habitats



A rendering of the commercial space station by Orbital Technologies



An illustration of what the Bigelow Commercial Space Station will look like in its final avatar station by Orbital Technologies